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Contract 5805

Field QC - Canobie AGG Survey

XCalibur MP Survey 902212

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for Geoscience Australia

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Summary

This short report describes the quality control checks performed on the point-located data from the Canobie AGG survey flown by XCalibur MP from 24 November, 2021 to 3 December, 2021 over 10 flights. The data were all passed. There were shortcomings in the channels supplied but these were judged too minor to warrant re-work.

0.0.1 Channels Supplied

There are two errors in the fields supplied but neither warrant re-flight or correction. Firstly, the `Time_Day` field is supposed to be seconds past midnight, local time but the values at the start of the first lines of a flight are negative, suggesting that they are from a different time zone. Secondly, there is no Customer Project Number field.

0.0.2 Aircraft Static Position

The static position repeat-point measurements must be the same to within 1 m horizontally and 2 m vertically.

For flights 1 to 10, the post-flight and pre-flight positions were the same as the first measured position to better than 0.48 m horizontally and 0.15 m vertically. Over all pre- and post-flight positions, the standard deviations were: X - 0.14 m, Y - 0.20 m, Z - 0.04 m.

0.0.3 Quiescent Noise

The quiescent noise must be less than $10.0\,\mathrm{E}^2$/rad/sec over the band 0.85 - 0.95 / 1.05 - 1.15 Hz. For flights 1 to 10, all quiescent noise figures were below $3.2\,\mathrm{E}^2$/rad/sec.

It was noted that A complement quiescent values were all higher than B complement values so signal powers from flight data were checked to ensure that this difference did not affect survey data. At very low turbulence, the A signal remained higher but at moderate to high turbulence the B signal was higher. After dynamic corrections the differences between complements entirely disappear demonstrating that the differences in static and survey performance are suitably removed in processing.

0.0.4 Image Inspection

All geophysical fields were gridded using minimum curvature and shaded images of each were visually inspected. No artefacts or other problems were observed.

The texture of G_{NE} was parallel to NE and NW; that of G_{UV} was parallel to N and E; and that of G_{DD} had no preferred direction: all exactly as expected. The ratio of the amplitude of these signals was consistent with expectations with G_{DD} having several times higher magnitude than the other two. The geology strikes closer to N than NE, and the magnitude of G_{UV} was consequently a little higher than G_{NE}, again as expected.

The images of A_{NE}, B_{NE} and G_{NE} were similar to each other as expected and similarly for A_{UV}, B_{UV} and G_{UV}.

0.0.5 Ground Speed

Ground speed not to be outside the range $62\,\mathrm{m\,s^{-1}} \pm 20\%$ for more than 1000 m. The phrasing here suggests that one should check the ground speed between all successive sample pairs and flag an error if the range is exceeded for all sample pairs over any 1000 m segment. My algorithm checks the average speed in a moving window of 1000 m and reports an error if the average speed is outside the range for any window.

For flights 1 to 10, there were no exceedances and all segments had speeds within the range [-16%, +8%] of the nominal value.

0.0.6 Survey Area Turns

Although it is not a formal specification, it is useful to check that no aircraft end-of-line turns were commenced or completed within the survey area. The `Bearing` field was mean-adjusted to zero for each line, and gridded using bi-directional gridding with zero interpolation between lines and minimal interpolation along lines. The imaged grid is intended to highlight changes in bearing along line and large changes at the end or beginning of a line indicate that part of a turn was within the survey area. No obvious problems with turns were observed.

0.0.7 GNSS Statistics

The HDOP must be less than 4 at all times. For flights 1 to 10, it was always less than 1.1.

The VDOP must be less than 4 at all times. For flights 1 to 10, it was always less than 3.6.

The PDOP must always be less than 4 at all times. For flights 1 to 10, it was always less than 3.7.

The number of GNSS satellites available must never be smaller than 5. For flights 1 to 10 the smallest number available was 7.

0.0.8 Data Gaps

Gaps in data should not exceed 0.5 s and there should be no more than 2 gaps of any size in any one line.

For flights 1 to 10, no gaps were found in any fields.

0.0.9 Horizontal Position

The difference between actual and planned horizontal position shall not exceed 40 m for more than 1000 m.

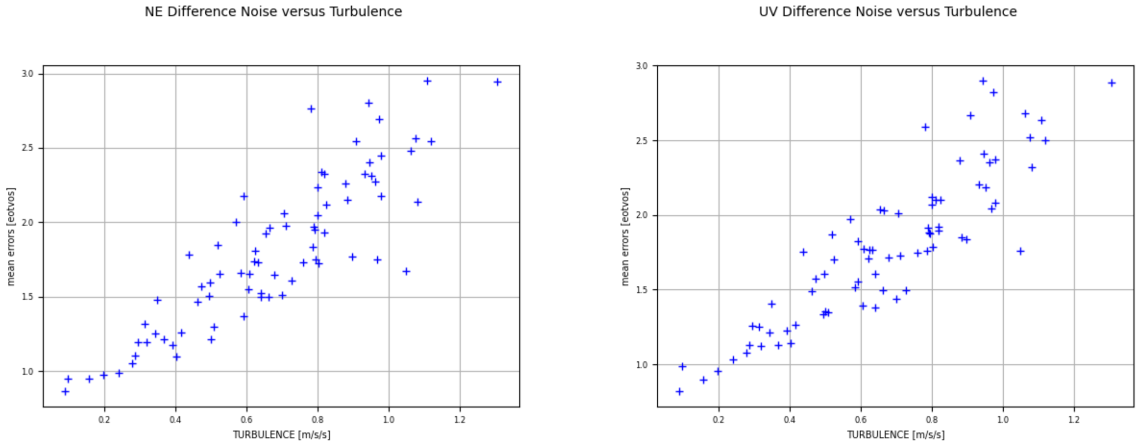

Figure 0.1: Plots of difference noise versus mean turbulence for G_{NE} (left) and G_{UV} (right). Each '+' represents the result for a single line in the survey.

For flights 1 to 10, all delivered lines successfully met this specification with the longest exceedance being about 120 m on line 100260. Some flown lines failed the specification but these were not delivered.

0.0.10 Vertical Position

The difference between actual and planned vertical position shall not exceed 20 m for more than 1000 m. In this case, planned vertical position is a ground clearance of 80 m.

For flights 1 to 10, all delivered lines successfully met this specification. Seventeen lines exceeded 20 m difference with the longest exceedance being about 450 m on line 100470; its peak difference of 38 m higher clearance was the largest single point difference.

0.0.11 AGG Difference Noise

The difference noise in both NE and UV components is required to be below 5.0 E RMS over a 0.18 Hz bandwidth for the entire line. This was checked using the fields (`ANE_TC_2p67`, `AUV_TC_2p67`, `BNE_TC_2p67`, `BUV_TC_2p67`). For flights 1 to 10, all lines had a noise below 3.0 E in mean turbulences up to 1.3 $\mathrm{m\,s^{-2}}$ (Fig 0.1).

The noise fields (`Noise_NE`, `Noise_UV`) were mean-adjusted to zero and gridded using bi-directional gridding with no interpolation across lines and minimal interpolation along lines. These grids are designed to make any bursts of unusually high noise along any line clearly visible. None were seen.

The difference noise data were plotted against turbulence and the plots compared with published results[1] for the same Falcon AGG sensor (Feynman) in the same aircraft type. The results were very

[1] D. B. Boggs, J. B. Lee, E. H. van Leeuwen, R. A. M. Maddever, R. J. Turner, M. Downey, G. Liu, and K. McCracken. The BHP-Billiton Digital Gravity Gradiometer. Preview, 117:21–22, Aug. 2005.

similar suggesting that the sensor is performing to expectations.

0.0.12 Survey Location

A grid of the vertical gravity gradient was exported to kml format and displayed in Google Earth Pro to confirm that the survey location was consistent with the specifications. This method is not of sufficient precision to reveal errors in datum but did show that the location was not in serious error.